They Didn't Know They Were Learning: an Intuitive Approach to Cybersecurity Education

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Abstract. What if the most effective way to teach a technical concept is to not teach it at all?
That question emerged from a practical observation, not a theoretical premise. A competition designed to teach security thinking to nineteen secondary schools produced something unexpected: students were engaging - but through pattern matching, not understanding. The activity worked on the surface and failed underneath. That gap between intended outcome and observed behaviour prompted a research inquiry that led to the Intuitive Cybersecurity Education (ICE) framework.
ICE is built on a single proposition: that a game whose internal mechanics are structurally identical to the real system being taught will build transferable, procedural knowledge - in learners who do not know they are being taught. The framework does not supplement instruction with games. It replaces instruction with consequential play, and allows knowledge to emerge from the doing rather than the telling.
Signal is the first game deliberately built on ICE principles. Players work inbound and outbound on a live relay strip - decoding survivor locations and, after the opening arc, encoding short acknowledgements - across binary, hexadecimal, ASCII, Base64, and URL representations because survival demands it, not because a lesson names the formats first. Player-facing relay copy avoids encoding jargon; vocabulary and worked examples belong in the debrief after play. The knowledge emerges. The instruction comes after.
The ICE framework is theoretically grounded in established learning science and designed to address the transfer failure that existing approaches - including well-intentioned game-based ones - consistently produce. Signal is its first deliberate implementation. Whether it achieves what theory predicts is the question this paper raises, and a planned controlled evaluation will answer.

Keywords: Cybersecurity education · Game-based learning · Data representation · Intuitive learning · Transfer of learning · Procedural knowledge · ICE framework.

1 Introduction

The cybersecurity skills gap is one of the most widely documented challenges in the digital economy. While previously characterized by a general shortage of per-

sonnel, the challenge is shifting, with organizations now prioritizing the critical need for specialized skills over a general headcount deficit[9], and the pipeline problem begins not in higher education but earlier - in the way foundational concepts are introduced to learners encountering the field for the first time.

Data encoding sits at the base of this conceptual stack. Binary representation, hexadecimal notation, ASCII character mapping, Base64 encoding, and URL encoding are not peripheral topics - they appear throughout network protocols, cryptographic systems, web application security, malware analysis, and digital forensics. A learner who does not have an intuitive grasp of encoding cannot fully understand how data is transmitted, stored, or obfuscated. Yet encoding is typically taught in one of two ways: as a set of conversion rules to be memorised, or as a worked example to be followed. Neither approach builds the flexible, transferable understanding that security work demands.

This paper presents Signal, a browser-based game that teaches all five encoding formats through play. Signal is built on the Intuitive Cybersecurity Education (ICE) framework, which holds that durable learning in technical domains occurs not through instruction but through the construction of correct mental models - and that games whose mechanics are structurally identical to the real systems they represent are a very efficient vehicle for that construction.

The paper is structured as follows. Section 2 reviews existing approaches to game-based learning in cybersecurity education. Section 3 describes the ICE framework and its design principles. Section 4 presents Signal's implementation. Section 5 reports early engagement evidence. Section 6 discusses implications and limitations. Section 7 outlines future work. Section 8 concludes.

2 Related Work

Game-based learning in cybersecurity education has attracted growing research attention, with approaches ranging from Capture the Flag (CTF) competitions[4] to purpose-built simulation environments[7] and tabletop exercises[13].

CTF competitions are the most widely adopted game-based format in cybersecurity education. They develop problem-solving skills and technical depth, and have been shown to increase learner engagement and motivation[4][11]. However, CTFs presuppose a baseline of technical knowledge and are poorly suited to introducing foundational concepts to novice learners. A student who does not yet understand what hexadecimal represents cannot benefit from a CTF challenge that requires hex manipulation.

Gamification of existing curricula - adding points, badges, and leaderboards to conventional instruction - has produced mixed results. Where the underlying instructional design is weak, gamification does not compensate for it[8]. Several studies have found that extrinsic reward structures can undermine intrinsic motivation when they are decoupled from the genuine challenge of the learning task[15].

Purpose-built educational games for cybersecurity include platforms such as CyberCIEGE[17], Countermeasures[10], and various commercially available

products. These typically model security decision-making at a systems level, which is valuable but does not address the conceptual foundations that make such decisions legible. While a robust body of literature explores game-based learning for high-level programming logic[12][5][16] and network security topologies[17][10], low-level data representation remains under-addressed. Existing digital games in this space focus almost exclusively on mathematical speed-drills for binary-to-decimal conversion[2][3]. Consequently, there is a notable gap in the literature for games that intuitively address the broader encoding pipeline - a gap that Signal is designed to fill.

The ICE framework draws on constructivist learning theory[6], which asserts that learners actively construct meaning and mental models by interacting with an environment, testing hypotheses, and observing the consequences of their actions, and on the concept of tacit knowledge[14], which distinguishes between knowledge that can be stated explicitly and knowledge that is held in the body as trained intuition. The goal of ICE is to engineer the construction of tacit, transferable cybersecurity knowledge - the kind of understanding that survives the absence of a reference sheet.

3 The Intuitive Cybersecurity Education Framework

The Intuitive Cybersecurity Education (ICE) framework is built on three design principles, the conceptual architecture of which was fundamentally inspired by pedagogical design methodologies outlined by Nic Barker in a presentation on intuitive education games [1]:

Principle 1: Structural identity between mechanic and system. The game mechanic must encode the logic of the real system, not merely its theme. A game that uses a "hacking" aesthetic to represent a slot machine teaches nothing about cybersecurity. A game in which the player's task requires the same cognitive operations as the real technical problem - grouping, mapping, converting - builds the same mental model. Structural identity is the necessary condition for knowledge transfer.

Principle 2: Subconscious knowledge transfer. Instruction makes learning visible and, in doing so, often makes it brittle. When a learner is told "this is how binary works," they engage declarative memory - encoding a rule. When a learner is required to decode a binary string to progress in a game, they engage procedural memory - building a skill. The ICE framework requires that no educational framing be visible to the player. The player is solving a game problem. The fact that this game problem has the same structure as a real technical problem is a discovery the player makes after the fact, not a premise they are given in advance.

Principle 3: Enjoyment as a prerequisite, not a reward. A game that is not genuinely enjoyable will not sustain engagement long enough for knowledge transfer to occur. The ICE framework treats enjoyment as a design constraint, not a motivational technique. Every game built on ICE must

pass a simple test: a player who has never heard of or is not interested in cybersecurity should enjoy it on its own terms. If the game only makes sense in an educational context, it fails the test.

These principles distinguish the ICE framework from both gamification (which adds game elements to instruction) and educational games (which use game aesthetics to deliver instruction). ICE does not deliver instructions at all. It engineers the conditions in which the correct mental model emerges from play. Figure 1 summarises the three principles and the play-then-debrief sequence.

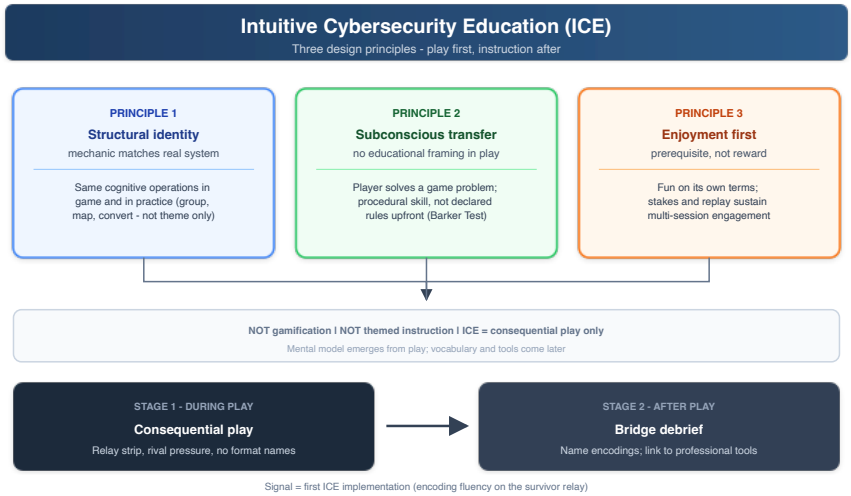


Fig. 1. The Intuitive Cybersecurity Education (ICE) framework: three design principles and the play-then-debrief sequence.

4 Signal: Design and Implementation

4.1 Narrative and Context Signal is set in the aftermath of the Cascadia Event - a planetary rupture that caused seven minutes of simultaneous tectonic movement, destroying every coastal city and rendering the surface uninhabitable. Billions died. Not everyone.

The player is the Signal Officer at Apex Station, a geoscience research facility built three kilometres underground into stable bedrock, designed to survive exactly this kind of event. 53 people made it to the emergency hatches. The player operates one of the few remaining working long-range communications array on Earth.

Survivors on the surface are transmitting through whatever device they have left - personal phones, emergency beacons, web forms, backup relay systems.

Every device encodes differently. Every signal arrives partially corrupted. The player runs a live relay shift on that stream: triage incoming pings under rival and budget pressure, decode inbound traffic to recover location and skills, and spend colony Ether on extractions when the roster plan allows — not authorise or deny rescue requests in a case file.

The stakes are real within the game world. Survivors that the player decodes too slowly, or fails to decode at all, are claimed by rival stations - other underground facilities that survived the Event and are competing to build the strongest colony. A survivor rescued becomes a named character contributing to New Geneva. A survivor missed is gone.

This narrative serves the ICE framework in three ways. First, it provides non-educational urgency: each signal is a person, not an exercise. Second, the multi-format encoding is narratively motivated - different devices on a destroyed surface use whatever protocol they have, which is why the player encounters binary radio pulses, compressed hex relay data, ASCII terminal output, Base64-encoded beacon payloads, and URL-encoded web form submissions. Third, the rival pressure introduces time as a game mechanic, creating the conditions under which encoding fluency develops through repetition under genuine stakes rather than drill.

4.2 Encoding Progression Signal teaches five encoding formats in sequence, each introduced as a natural evolution of the game's challenge. Inbound decoding of each format is introduced in the opening five signals; outbound encoding on the same representation is learned afterward, when the player must clear a short acknowledgement on the relay before a rescue can move. Figure 2 shows SIGNAL 001 during this arc: the player performs bit-grouping operations on the relay strip while format names remain off the tape.

Binary. The earliest transmissions arrive as raw binary streams. Players must group bits into bytes and convert them to reveal the hidden content. The game does not explain binary; it presents a pattern and requires the player to find it. The grouping mechanic - separating a continuous stream into discrete units of eight - encodes the fundamental structure of byte representation.

Hexadecimal. Later transmissions note that encoding the same message in binary takes up too many characters and risks interception by pattern-length analysis. The game introduces a more compact notation: hexadecimal. Players discover that two hex digits represent one byte - the same information, less space. The compression is experienced as a practical advantage, not explained as a fact.

ASCII Once players can decode hex, transmissions begin including ASCII-mapped content - text encoded as numeric values. Players build the mapping between numeric representations and characters through repeated exposure, discovering the ASCII table as a consequence of decoding rather than memorising it in advance.

Base64 Transmissions requiring binary data to be transmitted safely over text-based channels introduce Base64 encoding. Players encounter the character-

istic padding characters and the expansion ratio (three bytes becoming four characters) as game constraints before they are named as technical properties.

URL Encoding The final format addresses transmissions embedded in web-based tracking systems, where certain characters must be percent-encoded for safe transmission. Players encounter percent-encoded strings and learn to decode them by pattern recognition and substitution.



Fig. 2. Signal during the opening relay arc (SIGNAL 001). The player decodes survivor traffic using the same bit-grouping operations as real encoding work; format names are withheld until the post-session debrief.

4.3 Technical Implementation Figure 3 shows the deployment stack used for school and lab access without local install: browser Phaser client, Node.js API, Redis session cache, and PostgreSQL durable state. Gameplay intents are validated server-side; progression is not stored in `localStorage`.

Signal runs in the browser on standard school hardware, including Chromebooks. Decoding inbound survivor traffic follows one unified flow on the strip: Extract (pull a valid span from the noisy tape), Convert (do the real place-value or bit math to a stamped decimal byte), and Lock (map that byte to a character on a full chart). Encoding outbound traffic uses the same three beats in reverse - Lock, Convert, Extract - so the player builds wire on an outbound strip after they have read it on the inbound strip. Wrong spans, sums, or chart picks are rejected in place with comms feedback; the game does not auto-complete the math for them. That split mirrors how representations work on the wire and is the behavioural contract the project is shipping behind feature flags.

The opening survivor stream introduces all five target representations across the first five scripted signals, decode-only: SIGNAL 001 (binary, Mara Solis),

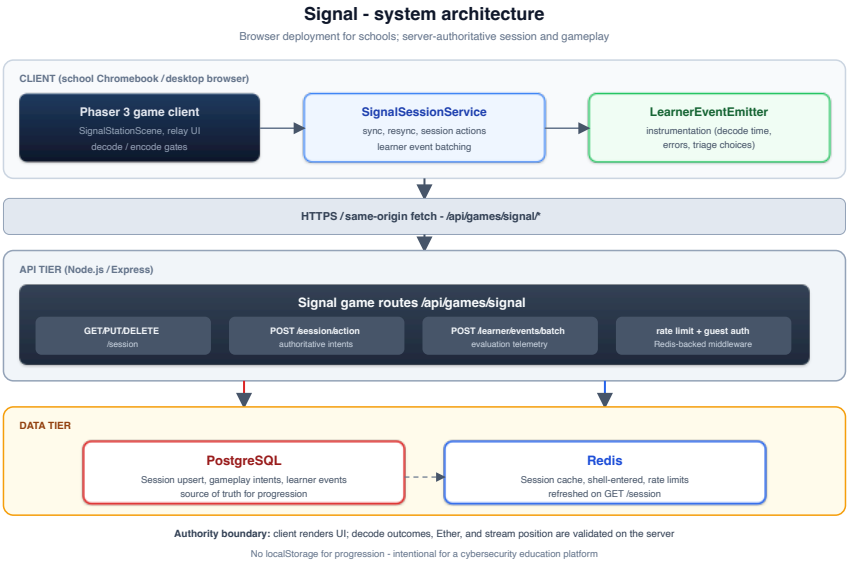


Fig. 3. Signal deployment architecture: browser-based Phaser client, Node.js API, Redis session cache, and PostgreSQL durable state. Gameplay intents are validated server-side.

SIGNAL 002 (hex), SIGNAL 003 (ASCII), SIGNAL 004 (Base64), and SIGNAL 005 (URL encoding). Each signal is a full inbound rescue beat under colony and rival stakes; per the design, outbound encoding does not appear in this intro band - the fiction is "read the stream first, then talk back". Post-intro play adds short outbound acknowledgement jobs on the same representation (decode a ping, then clear an ack on the outbound strip). Broader stream, economy, and colony systems remain in active development ahead of the conference demo.

Production rollout is phased: inbound gates are landing encoding by encoding in the main client; outbound gates follow the same pattern. The demonstration at Advances in Teaching and Learning for Cyber Security Education 2026 will use the most stable July build; attendees will at minimum work through the five-signal opening decode arc, and - where the build allows - a post-intro outbound ack that inverts the same mechanic.

5 Early Evidence

5.1 University Lab Context Signal has been informally introduced in a cybersecurity lab environment at UWE Bristol as a supplementary tool for a testbed of voluntary students encountering data representations for the first time. Observation during these sessions indicates that students who engage with the game's binary levels demonstrate greater fluency with subsequent related exercises than

those who encountered the same content through conventional instruction alone. A structured evaluation is planned for the next academic year.

5.2 Cyber Battleships: The Observation That Prompted ICE The ICE framework did not emerge from theory. It emerged from a problem observed in practice.

Cyber Battleships was a standalone competition designed to teach traffic analysis to secondary school students through a game-based challenge - nineteen schools participated. Teams read server log files to identify leaked data representing coordinates of potential attacks to a network and sink ships representing the attackers on a grid. The competition was not designed as an ICE implementation; the ICE framework did not yet exist.

What was observed during and after the competition was a disconnect between intended outcome and actual behaviour. Students were engaging with the challenge through pattern matching - recognising surface features of the encoded data rather than developing the security thinking the activity was designed to build. The game produced participation; it did not reliably produce the underlying conceptual understanding that was the goal.

That gap - between what a game-based activity intended to teach and what it actually produced - prompted the research inquiry that led to the ICE framework. Post-competition feedback gathered offered qualitative signals consistent with this observation: teachers noted that students engaged persistently with the competition challenge. These observations informed the research question ICE was built to answer - not as evidence of ICE's effectiveness, but as the origin of the problem it was designed to solve.

6 Discussion

6.1 Implications for Curriculum Design The ICE framework suggests a reordering of the relationship between intuition and instruction in cybersecurity education. Conventional curricula often introduce notation before application. ICE reverses that order on the relay: learners meet each representation as survival traffic first - decoding inbound wire in the opening five signals, then encoding outbound wire once they can read the stream - while formal vocabulary belongs in the debrief, not on the tape.

For data representation specifically, this has practical implications for how the topic is sequenced in cybersecurity programmes. Binary and hexadecimal are still often taught as abstract notation before students see them in protocols or logs. Signal's opening stream is built the other way around - five inbound encodings in five live signals - but the informal UWE lab observations reported in Section 5.1 are exploratory only. Whether application-first sequencing, including outbound encoding after inbound mastery, produces more durable understanding than conventional instruction is an open question; a planned controlled evaluation, with pre/post assessments and a comparison arm, is intended to answer it.

6.2 Generalisability Signal addresses data representations, but the ICE framework is not specific to this topic. The same design principles - structural identity, subconscious transfer, enjoyment as prerequisite - apply to any domain where the target knowledge has a clear logical structure that can be encoded in a game mechanic. Authentication, Network Routing, and Log Analysis all have clear structural analogues in game design. Signal is the first of a planned series of games applying the ICE framework across the foundational cybersecurity curriculum.

6.3 Limitations The evidence presented here is preliminary. The university lab observations are informal and lack a control condition. The school competition data is qualitative and gathered from a self-selected sample of respondents. A rigorous evaluation - randomised assignment, validated knowledge assessments pre and post intervention, comparison with conventional instruction - is required before claims about learning outcomes can be made with confidence. The evaluation programme outlined in Section 7 is intended to supply that evidence.

7 Future Work

The ICE framework and Signal implementation described here are design contributions; evidence on learning outcomes remains to be gathered systematically. The next phase will test whether application-first relay play followed by a bridge debrief produces durable encoding fluency compared with conventional instruction.

Evaluation is planned in three strands. First, a controlled study with UWE Bristol learners will use pre/post assessments and, where feasible, a comparison arm, extending the informal lab observations reported in Section 5.1. Second, feedback will be collected from educators and practitioners who interact with Signal at conference demonstrations and outreach events, using a short evaluation questionnaire. Participants will be asked for informed consent before their responses are used for research and product development. Third, server-side learner telemetry already instrumented in Signal (decode time, error patterns, triage choices) will be analysed alongside questionnaire data to relate reported experience to in-game behaviour.

The same questionnaire and consent process will be reused at future engagement events so that evidence can accumulate across settings rather than from a single session. Findings from these strands will inform refinement of the ICE framework.

8 Conclusion

This paper has presented the Intuitive Cybersecurity Education (ICE) framework and Signal, its first deliberate implementation, as a response to an observed

failure mode in game-based cybersecurity education: engagement without transferable understanding.

The central contribution is a design language, not a single level or conversion drill. ICE requires structural identity between game mechanic and real encoding work, subconscious transfer through play without front-loaded vocabulary, and enjoyment as a design constraint rather than an extrinsic reward. These principles combine in a play-then-debrief sequence that separates consequential play on the relay strip from formal vocabulary afterward.

Signal instantiates that language on a live survivor stream: one inbound mental model (Extract, Convert, Lock) and its outbound inverse (Lock, Convert, Extract), applied across binary, hexadecimal, ASCII, Base64, and URL representations before format names appear in a debrief. Browser deployment with server-authoritative session state makes the tool viable for school and lab contexts without local installation.

Early evidence is intentionally preliminary. Cyber Battleships supplied the problem statement; informal UWE lab sessions suggest improved fluency on follow-on exercises among voluntary participants; neither constitutes proof of ICE effectiveness. Claims about learning outcomes must wait on the structured evaluation programme outlined in Section 7.

Taken together, the framework, implementation, and evaluation plan offer a reproducible approach for teaching data representation through consequential play - and a testable hypothesis that instruction after intuitive mastery may produce more durable procedural knowledge than notation-first curricula.

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